

Databricks Certified Machine Learning Associate

Practice Exam — Questions & Answer Key (48 questions)

Correct answers are highlighted in green. Items marked $\triangle$ are worth double-checking (and three questions weren't captured).

Question 1

A data scientist is hyperparameter tuning a RandomForestClassifier using scikit-learn's GridSearch with `cross_validation=5` and a `param_grid` of: `max_depth [5,10,15,20]`, `max_features ['auto','sqrt','log2']`, `n_estimators [200,500]`. How many models will be trained in total?

- A. 120
- B. 24
- C. 45
- D. 14

Answer: A — 120.

Why: $4 \times 3 \times 2 = 24$ parameter combinations, each fit 5 times for CV $\rightarrow 24 \times 5 = 120$ model fits.

Question 2

A data scientist plans to use MLflow log functions to save various pieces of evidence. Which function can save plots that visualize key performance metrics?

- A. `mlflow.log_artifact()`
- B. `mlflow.log_param()`
- C. `mlflow.log_model()`
- D. `mlflow.log_metric()`

Answer: A — `mlflow.log_artifact()`.

Why: Plots are saved as files/artifacts; `log_artifact()` logs arbitrary files such as image plots.

Question 3

A data scientist must tune hyperparameters of a complex model on a moderately large dataset with a limited compute budget, has prior knowledge of likely ranges, and wants to efficiently explore without evaluating every combination. Which search strategy meets their needs?

- A. Random search of all hyperparameter combinations
- B. Point search of hyperparameter combinations selected from previous evaluations
- C. **Bayesian optimization on hyperparameters selected from previous evaluations**
- D. Grid search of all hyperparameter combinations within a specified range

Answer: C — Bayesian optimization.

Why: Bayesian optimization uses prior results to choose promising points, efficient under a limited budget.

Question 4

A column (number of customer service calls) has missing values and a highly skewed distribution (most 0–2 calls, a few very high). Which method correctly imputes the missing values?

- A. **Impute with the median value of the column**
- B. Impute with the mean value of the column
- C. Impute with a random sample from the existing values
- D. Impute with the mode value of the column

Answer: A — Impute with the median.

Why: The median is robust to skew/outliers, unlike the mean.

Question 5 verify

A house-price dataset has Neighborhood (categorical, 50 unique values) and House type (categorical, 4 values). The engineer must prepare categoricals for different model types. Which technique will do this?

- A. For linear regression, Neighborhood left as-is, while House type converted to integers
- B. For a random forest, both Neighborhood and House type should be one-hot encoded
- C. For linear regression, House type should be one-hot encoded, while Neighborhood (high cardinality) might benefit from a different encoding technique**
- D. For both linear regression and random forest, Neighborhood and House type should be one-hot encoded

Answer: C — One-hot House type; use a different encoding for high-cardinality Neighborhood.

Why: Low-cardinality categoricals suit one-hot; 50-level Neighborhood needs target/other encoding to avoid huge sparse vectors.

Question 6

A model to forecast power production is logged in Unity Catalog. The engineer must load the latest version and forecast over the next five days; data arrives in TB-scale batches and results must be produced at 15-minute intervals, as efficiently as possible. Which approach will do this?

- A. Load data as a streaming DataFrame, deploy the model to Model Serving, and use a UDF to call the endpoint per data point
- B. Load data as a pandas DataFrame and load the model from mlflow as a pyfunc model
- C. Load data as a Spark DataFrame, load the model from mlflow as a pyfunc model, and use a Pandas UDF to perform the forecasting**
- D. Load data as a pandas DataFrame and load the model from mlflow as a scikit-learn model

Answer: C — Spark DataFrame + mlflow pyfunc + Pandas UDF.

Why: Distributed batch scoring of TB-scale data needs Spark with a Pandas UDF; pandas-only doesn't scale.

Question 7 verify

A retail data scientist wants to predict next year's gross margin per country using AutoML on a UC dataset (GDP, inflation, unemployment, revenue, etc.). Which statement describes how AutoML facilitates feature selection?

- A. AutoML randomly selects features without specific criteria
- B. AutoML uses a brute-force method testing all possible feature combinations
- C. AutoML allows adjustment of feature selection based on feature importance scores, which must be enabled as an option and needs the notebook to re-run**
- D. AutoML relies solely on domain expertise to manually select features

Answer: C — Feature selection can be tuned via feature-importance scores (enabled as an option).

Why: AutoML supports importance-based feature selection that you can enable and re-run.

Question 8

In a house-price linear regression, prices are heavily skewed and residuals show a clear pattern (violating homoscedasticity). Which action accounts for these problems?

- A. Apply a square root transformation to the independent variables to achieve linearity
- B. Use a polynomial regression model to capture non-linear relationships
- C. Apply a log transformation to the dependent variable (house prices) to reduce skewness and improve model assumptions**

D. Remove the outliers from the dataset

Answer: C — Log-transform the dependent variable.

Why: Log-transforming the skewed target reduces skew and stabilizes variance (helps homoscedasticity).

Question 9

A data scientist created a Data Profile from the notebook result cell for EDA on a Spark DataFrame df. What is the programmatic way to accomplish the same task?

- A. df.info().show()
- B. dbutils.data.summarize(df)**
- C. dbutils.data.profile(df)
- D. df.describe().show()

Answer: B — dbutils.data.summarize(df).

Why: dbutils.data.summarize() generates the same data-profile summary programmatically.

Question 10

(Screenshot for Question 10 was not captured in extractable text.)

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Question 11

A team discusses how to achieve different ModelOps tasks in Databricks. Which approach is a correct use of Databricks features?

- A. Use AutoML to monitor model performance
- B. Use Unity Catalog to version-control model code
- C. Use Delta Live Tables for no-code model development
- D. Use MLflow to track model development**

Answer: D — Use MLflow to track model development.

Why: MLflow is the tracking/experiment tool; the others misassign each product's purpose.

Question 12

A dataset has customer_category (3 levels) and preferred_payment_method (4 levels). The data scientist wants to determine if there is a significant relationship between these two categorical variables. Which statistical test is appropriate?

- A. Chi-Square Test of Independence**
- B. One-way ANOVA
- C. Two-sample t-test
- D. Pearson correlation coefficient

Answer: A — Chi-Square Test of Independence.

Why: The chi-square test assesses association between two categorical variables.

Question 13

(Screenshot for Question 13 was not captured in extractable text.)

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Question 14

A data scientist will evaluate a model on unseen data; a senior suggests cross-validation instead of a train-validation split. Which statement is a benefit of a train-validation split?

- A. It provides more reliable estimates for model performance
- B. It uses all available data for training and testing
- C. It generates more metrics to assess model performance
- D. It runs faster and requires less processing time**

Answer: D — It runs faster and requires less processing time.

Why: A single split trains once (vs k times for CV), so it's faster but less robust.

Question 15

A data scientist wants to programmatically identify the best run from an MLflow experiment by highest accuracy using the MLflowClient API (experiment_id given). Which code block achieves this?

- A. `client = MlflowClient()` (no search)
- B. `best_run = client.get_best_run(experiment_id, metric='accuracy')`
- C. `best_run = client.search_runs(experiment_id, order_by=["metrics.accuracy DESC"], max_results=1)[0]`**
- D. `runs = client.search_runs(experiment_id); best_run = min(runs, key=lambda r: r.data.metrics['accuracy'])`

Answer: C — `search_runs(..., order_by=["metrics.accuracy DESC"], max_results=1)[0]`.

Why: `order_by` accuracy DESC with `max_results=1` returns the highest-accuracy run; `get_best_run` isn't a real method, and `min()` returns the worst.

Question 16 verify

Which TWO statements accurately describe information available on specific pages within the MLflow UI in Databricks? (Choose 2)

- A. The Run Page includes a Model tab that shows the model's input schema and sample input
- B. The Artifacts Page allows users to upload and manage datasets used for model training
- C. The Compare Runs Page enables visualizing differences in hyperparameters across runs using parallel coordinates plots**
- D. The Experiment Run Page displays a graph of model accuracy over time during training epochs
- E. The Experiment Page shows a table of runs with information about the dataset, source code notebook, and any created models**

Answer: C, E — Compare Runs parallel-coordinates plot (C) and the Experiment Page runs table (E).

Why: C and E describe real MLflow UI features; B and D are false, and A's 'Model tab/sample input' phrasing is the weaker claim.

Question 17

A data scientist optimizes a RandomForestClassifier with Hyperopt's fmin (objective returns {'loss': -accuracy}, algo=tpe.suggest, trials=SparkTrials()). Which statement about this implementation is correct?

- A. SparkTrials() serializes the trials but won't parallelize the hyperparameter search
- B. The objective minimizes accuracy, so the best model will have the lowest accuracy
- C. `hp.choice` is for continuous hyperparameters while `hp.uniform` is for discrete ones
- D. fmin will return the best hyperparameters that minimize the loss (maximize the accuracy)**

Answer: D — `fmin` returns the hyperparameters that minimize loss (i.e., maximize accuracy).

Why: Loss = -accuracy, so minimizing loss maximizes accuracy; SparkTrials parallelizes, and `hp.choice` is discrete.

Question 18

An engineer wants to use `FeatureEngineeringClient.score_batch` to score new data against a Unity Catalog model for batch inference. What requirement must they consider?

- A. The model requires manual feature engineering to look up feature values
- B. The input data must include all feature values, including those that can be looked up from feature tables
- C. The model can only be used for training, not inference on new data
- D. The input data requires the lookup key column and the method will automatically look up all feature values from the feature tables**

Answer: D — Provide the lookup key; `score_batch` auto-looks up features from feature tables.

Why: With a packaged feature-lookup model, you supply keys and the client retrieves the feature values.

Question 19

A categorical feature 'color' has values 'red','blue','green'. Which approach correctly implements one-hot encoding for the color feature using PySpark?

- A. Use Spark's built-in functions to one-hot encode the color column directly
- B. Apply `pd.get_dummies` after converting the Spark DataFrame to pandas
- C. Use `StringIndexer`, then `OneHotEncoder`, then `VectorAssembler` to create the final feature vector**
- D. Apply the `categorical_column` function, then the `OneHotEncodingEstimator`

Answer: C — `StringIndexer` → `OneHotEncoder` → `VectorAssembler`.

Why: Spark ML one-hot encoding requires indexing the strings first, then encoding, then assembling features.

Question 20

A financial company needs: Daily Financial Reports (end-of-day, large volumes), Fraud Detection (immediate, as transactions occur), and Customer Behavior Analysis (near real-time). Which combination of data-processing approaches meets all requirements cost-effectively?

- A. Real-time for daily reports, batch for fraud, stream for customer behavior
- B. Batch for daily reports, stream for fraud, real-time for customer behavior
- C. Batch for daily financial reports, real-time processing for fraud detection, and stream processing for customer behavior analysis**
- D. Stream for daily reports, batch for fraud, real-time for customer behavior

Answer: C — Batch (daily reports) + real-time (fraud) + stream/near-real-time (customer behavior).

Why: End-of-day reports suit batch; fraud needs real-time; near-real-time behavior fits streaming.

Question 21

A data scientist chose an offline feature store (over online) for features used primarily for batch processing. What aspect of offline feature stores enabled this decision?

- A. They prioritize low-latency access optimized for real-time point lookups (key-value stores)
- B. They support real-time and batch equally with best performance for all use cases
- C. They are optimized for batch processing and efficiently handle large volumes of data, often using data lakes or data warehouses**
- D. They only support batch processing, making them the only viable option for non-real-time apps

Answer: C — Optimized for batch processing of large data volumes (lakes/warehouses).

Why: Offline stores are built for high-throughput batch retrieval over large datasets.

Question 22

Using the Databricks Feature Store with `FeatureEngineeringClient` instantiated, what are the minimum parameters required for the `write_table` function (params available: `name`, `df`, `mode`, `primary_key`)?

- A. **name, df, mode**
- B. name, df
- C. df
- D. name, df, primary_key

Answer: A — name, df, mode.

Why: `write_table` requires the target table name, the `DataFrame`, and the write mode.

Question 23

An ML engineer queries a model deployed to a real-time Model Serving endpoint (created/deployed by a teammate) using a PAT, but gets an error. Why is this happening?

- A. The engineer does not have EXECUTE privileges on the registered model in Unity Catalog
- B. The engineer does not have CAN VIEW permissions on the endpoint
- C. Model serving endpoints can only be queried using OAuth tokens
- D. **The engineer does not have CAN_QUERY permissions on the endpoint**

Answer: D — Missing `CAN_QUERY` permission on the endpoint.

Why: Querying a serving endpoint requires the `CAN_QUERY` permission on that endpoint.

Question 24

An engineer wants streaming inference on customer-churn data using Delta Live Tables; the model is a SparkML pipeline saved in MLflow and new data streams into a Delta table. Which approach does this?

- A. Load the MLflow model with `mlflow.pyfunc.load_model()` and apply it row-by-row inside a streaming DLT function via `dlt.read()`
- B. Use `mlflow.sklearn.load_model()`, read with `spark.readStream()`, convert each batch to pandas, apply with `df.apply()`, write with `spark.writeStream()`
- C. Download the MLflow artifacts to DBFS and define a Spark ML Pipeline inside the DLT function over `dlt.read_stream()`
- D. **Create a Spark UDF from the MLflow model with `mlflow.pyfunc.spark_udf()`, then in a streaming DLT function use `dlt.read_stream()` and apply the UDF for predictions**

Answer: D — Create a `spark_udf` from the model and apply it over `dlt.read_stream()`.

Why: `mlflow.pyfunc.spark_udf` gives a distributed UDF that applies cleanly to a streaming DLT source.

Question 25

A data scientist created a feature store table; UC is now enabled. What is a primary benefit of creating feature store tables in Unity Catalog vs directly in the workspace?

- A. Feature tables in UC provide improved performance
- B. Feature tables in UC can reduce storage costs
- C. **Feature tables in UC are automatically available in any workspace that has access to the catalog**
- D. Feature tables in UC are automatically encrypted

Answer: C — They're available in any workspace with access to the catalog.

Why: UC feature tables are governed centrally and shareable across workspaces bound to the metastore.

Question 26

For a daily-sales time-series forecast, the feature set has Product Category (categorical, missing at random), Sales Amount (skewed, missing at random), and Transaction Count (normally distributed with a few out-of-range outliers, missing at random). Which imputation strategy is most likely to improve quality?

- A. Product category use mode; sales amount use median (per month); transaction count use mean (per month)**
- B. Product category use mode; sales amount and transaction count use mean (per month)
- C. Remove all out-of-range and missing values from the training set
- D. Product category use mode; sales amount use mean (per month); transaction count use median (per month)

Answer: A — Mode (category), median (skewed sales), mean (normal transaction count).

Why: Median suits skewed Sales Amount; mean suits the roughly-normal Transaction Count; mode suits the category.

Question 27

Consider: `assembler = VectorAssembler(inputCols=[...], outputCol="features"); lr = LogisticRegression(maxIter=10, regParam=0.01)`. Which statement is true about this code?

- A. Both `VectorAssembler` and `LogisticRegression` are transformers
- B. `VectorAssembler` is an Estimator and `LogisticRegression` is a Transformer
- C. `VectorAssembler` is a Transformer and `LogisticRegression` is an Estimator**
- D. Both `VectorAssembler` and `LogisticRegression` are estimators

Answer: C — `VectorAssembler` is a Transformer; `LogisticRegression` is an Estimator.

Why: `VectorAssembler` transforms columns without fitting; `LogisticRegression` must be fit (estimator).

Question 28

A data scientist deployed Model A and Model B for real-time inference via Model Serving and wants A/B testing routing 70% to A and 30% to B, as easily as possible. Which approach will do this?

- A. Create a single serving endpoint and use the traffic split feature to allocate 70% to A and 30% to B**
- B. Use Delta Live Tables to stream incoming data and split it before sending to the models
- C. Deploy each model to a separate endpoint and implement custom client routing logic
- D. Use MLflow's built-in A/B testing functionality to automatically split requests

Answer: A — Single endpoint with traffic split (70/30).

Why: Model Serving's traffic-split feature serves multiple model versions from one endpoint by percentage.

Question 29 verify

A data scientist runs scikit-learn `GridSearchCV` on Databricks; it's slow and the cluster is largely unused (most work on the driver). They want to better leverage the cluster without rewriting all code, minimizing manual effort, and log every combination in MLflow. Which approach will do this?

- A. Switch to 'MLflow's `GridSearchCV`', which auto-distributes training across executors
- B. Use specialized libraries such as Hyperopt with SparkTrials or Optuna with Joblib**
- C. Switch from scikit-learn/pandas to PySpark with Spark ML and use Spark ML's `CrossValidator`
- D. Use widgets to parameterize training and a Databricks Workflows job calling the notebook multiple times on single-node clusters

Answer: B — Use Hyperopt+SparkTrials (or Optuna+Joblib) to distribute and log to MLflow.

Why: These libraries distribute tuning across the cluster and integrate with MLflow with minimal rewrite (no 'MLflow GridSearchCV' exists).

Question 30

A data scientist wants to understand the advantages of Databricks ML runtimes over standard runtimes. Which aspect is a key advantage of using ML runtimes?

- A. They automatically generate and optimize ML models without human intervention
- B. They provide an optimized environment for model development and deployment, ensuring reproducibility and easier management of dependencies**
- C. They eliminate the need for data preprocessing and feature engineering
- D. They protect against model bias and drift by continuously retraining

Answer: B — An optimized, reproducible environment with managed ML dependencies.

Why: ML Runtime preinstalls and manages ML libraries for a consistent, reproducible environment.

Question 31 verify

A data scientist needs to replace incorrect tag keys on model versions of a model registered in Unity Catalog. What are the minimum permissions required?

- A. UPDATE TAG privilege on the registered model, USE SCHEMA, and USE CATALOG
- B. CREATE_TAG and DELETE_TAG privilege on the registered model, USE SCHEMA, and USE CATALOG
- C. APPLY TAG VERSION privilege on the registered model, USE SCHEMA, and USE CATALOG
- D. APPLY TAG privilege on the registered model, USE SCHEMA on its schema, and USE CATALOG on its catalog**

Answer: D — APPLY TAG on the model + USE SCHEMA + USE CATALOG.

Why: Setting/replacing tags needs APPLY TAG on the object plus USE on the parent schema and catalog.

Question 32

A practitioner wants to visually check the linearity assumption (between independent and dependent variables) for a linear regression. Which visualization technique shows linearity?

- A. Plot each independent variable against other independent variables
- B. Plot the average of the dependent variable vs. the residual of the predicted value
- C. Plot the residuals of the predicted variable vs. the actual dependent variable
- D. Plot the independent variable against the dependent variable and visually inspect for linearity**

Answer: D — Plot the independent variable against the dependent variable.

Why: A scatter of X vs. Y directly reveals whether their relationship is linear.

Question 33

A fraud-detection dataset is highly imbalanced (fraud = 2%), causing high accuracy but poor recall. Which strategy should they implement to mitigate the class imbalance in the training data?

- A. Use ML runtimes to add interactive visualizations and fix imbalance in the notebook
- B. Use a feature selection method such as Shapley to remove irrelevant features
- C. Implement a cross-validation technique to ensure good generalization
- D. Apply a resampling technique such as SMOTE to increase the number of fraudulent samples**

Answer: D — Apply SMOTE (resampling) to balance the classes.

Why: SMOTE synthesizes minority-class samples, directly addressing the imbalance.

Question 34

A data scientist wants to accelerate model development by automating a routine task. Which task is typically automated by AutoML in Databricks?

- A. Deploying the model on a serving endpoint
- B. Creating a feature engineering set for model development
- C. Training and optimizing different models**
- D. Monitoring model performance for inference

Answer: C — Training and optimizing different models.

Why: AutoML automatically trains and tunes multiple models and surfaces the best.

Question 35

A model identifies fraudulent credit-card transactions and must prioritize catching as many frauds as possible (fraud $\approx$ 0.2%; fraud is the Positive class). Which primary evaluation metric should be used for model selection?

- A. Accuracy
- B. Recall**
- C. Precision
- D. R2

Answer: B — Recall.

Why: Catching the most fraud means maximizing true positives among actual frauds — recall.

Question 36

A dataset of numerical data is assumed normally distributed with a few extreme outliers to identify. Which outlier-detection approach will do this?

- A. Calculate the IQR; data below ($Q1 - 1.5 \times IQR$) or above ($Q3 + 1.5 \times IQR$) indicates extreme outliers**
- B. Run `.summary()` and observe an 'outlier' attribute equal to 1
- C. Run `.describe()` and observe an 'outlier' attribute equal to 1
- D. Use `max()/min()` to find points in the top/bottom 1%

Answer: A — Use the IQR rule ($Q1 - 1.5 \times IQR$, $Q3 + 1.5 \times IQR$).

Why: The $1.5 \times IQR$ fences are the standard statistical method to flag outliers; `.summary()/describe()` have no 'outlier' attribute.

Question 37

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Question 38

A stock-price model applied a log transformation on the dependent variable to reduce skewness. To estimate the impact of the independent variable on the response, which step is necessary before interpreting the results?

- A. Nothing in particular
- B. Exponentiate the coefficient**
- C. Exponentiate the independent variable
- D. Divide the coefficient by 100

Answer: B — Exponentiate the coefficient.

Why: With a log-transformed target, coefficients are interpreted on the original scale by exponentiating them.

Question 39

A data scientist will train a customer-churn model that runs in batch daily; features already exist in a UC feature table. Which TWO are advantages of using the Databricks Feature Engineering Client to train the model? (Choose 2)

- A. Automatically passes model lineage information back to Unity Catalog**
- B. The model will automatically inherit feature values when later running inference**
- C. Faster compute engine than Spark
- D. It is specialized for batch processing models only
- E. The package can read domain-specific column names to create relevant features

Answer: A, B — Auto-records lineage to UC (A) and packages feature lookups for inference (B).

Why: *The client logs feature lineage and bundles lookups so inference auto-retrieves feature values.*

Question 40

A house-price model uses a loss function that heavily penalizes large errors; training struggles to converge with unstable predictions. Why is it crucial to select the right loss function here?

- A. The loss function controls the size of the training set
- B. The loss function ensures that the data is balanced
- C. The loss function determines the computational efficiency of the model
- D. The loss function directly influences how the model's parameters are updated during training**

Answer: D — It directly influences how parameters are updated during training.

Why: *The loss defines the gradients that drive parameter updates, affecting convergence and stability.*

Question 41

Comparing two deployment patterns — 'deploy model' (promote the model artifact) vs 'deploy code' (promote the training code) — which scenario is INAPPROPRIATE for the deploy-model approach?

- A. Production data is not accessible from the development environment**
- B. All work is done in a single Databricks workspace
- C. Model training is very expensive or hard to reproduce
- D. The data scientist is not working with external repos or a CI/CD process

Answer: A — When production data isn't accessible from dev.

Why: *Deploy-model trains in dev then promotes the artifact; if prod data isn't reachable from dev, you can't train properly — deploy-code fits better.*

Question 42 verify

After A/B testing, a model registered in Unity Catalog has version 2 (current production) and version 3 (new, proven better). What is the correct approach for switching the production version?

- A. On the MLflow Model Registry UI set version 2 to Archived and version 3 to Production (stages)
- B. Call `transition_model_version_stage()` for both versions (Archived / Production)
- C. Call `set_registered_model_alias()` to reassign the new version as the 'Champion' alias (e.g., `set_registered_model_alias('prod.ml_team.my_model','Champion',3)`)**
- D. In the catalog UI, change a 'status' tag for v2 to Archived and v3 from Challenger to Champion

Answer: C — Use `set_registered_model_alias()` to point 'Champion' at version 3.

Why: *UC model registry uses aliases (not stages); reassigning the Champion alias promotes the new version.*

Question 43

A logistic-regression sklearn pipeline currently handles only numeric features (SimpleImputer mean + StandardScaler). The data scientist wants to handle both numeric and categorical features. Which code block does this?

- A. A single Pipeline with StandardScaler, OneHotEncoder(drop='first'), and LogisticRegression
- B. A ColumnTransformer with a numeric Pipeline (SimpleImputer mean + StandardScaler) on numeric_features and a categorical Pipeline (SimpleImputer most_frequent + OneHotEncoder) on categorical_features, then a Pipeline with the classifier**
- C. Two separate pipelines run independently and concatenated manually
- D. Apply OneHotEncoder to all columns then LogisticRegression

Answer: B — Use a ColumnTransformer combining numeric and categorical sub-pipelines.

Why: ColumnTransformer applies the right imputation/encoding per feature type before the classifier.

Question 44

A text-classification sentiment model must detect and categorize both positive and negative feedback with equal precision (balanced, unbiased). Which evaluation metric is best suited?

- A. RMSE
- B. F1 Score**
- C. Recall
- D. Precision

Answer: B — F1 Score.

Why: F1 balances precision and recall, suiting a balanced assessment across both classes.

Question 45

A model was trained and logged to MLflow; the data scientist wants to register it in Unity Catalog (UC-enabled workspace) into the sales catalog and churn schema, in the same notebook. What is the correct way?

- A. `mlflow.set_registry_uri('databricks-uc');` `infer/set signature;` `mlflow.register_model(model_uri, 'churn_prediction_model')`
- B. `infer/set signature;` `mlflow.register_model(model_uri, 'churn_prediction_model')` (no registry URI)
- C. `mlflow.set_registry_uri('databricks-uc');` `infer/set signature;` `mlflow.register_model(model_uri, 'sales.churn.churn_prediction_model')`**
- D. register with a two-level name and no registry URI

Answer: C — Set registry URI to 'databricks-uc' and register with the 3-level name `sales.churn.<model>`.

Why: UC registration requires the databricks-uc registry URI and a fully-qualified `catalog.schema.model` name.

Question 46

A data scientist collaborates with other organizations that have their own Databricks accounts/workspaces in different regions and must decide between the Unity Catalog registry and the workspace registry for cross-organization collaboration. Which UC model registry capability enables this?

- A. Model sharing using Delta Sharing**
- B. Model discoverability across workspaces
- C. Centralized access control
- D. Model auditing and lineage

Answer: A — Model sharing using Delta Sharing.

Why: *Delta Sharing lets UC-registered models be shared securely across accounts/regions.*

Question 47

A random forest classifier performs exceptionally on training data but predicts essentially randomly on held-out test data. Which reason for the poor results is most plausible?

- A. The model is too simple, with a small number of very shallow trees
- B. The model is too complex, with a large number of very deep trees**
- C. The model needs k-fold cross-validation instead of a held-out dataset
- D. The model was trained on too much data

Answer: B — Overfitting — too complex (many very deep trees).

Why: *Great train but poor test performance is classic overfitting from overly deep/complex trees.*

Question 48

An engineer evaluating a linear regression model considers RMSE, MAE, R-squared, etc. Which statement is correct?

- A. A higher value of RMSE is considered desirable
- B. A higher value of R-squared is considered desirable**
- C. A higher value of MSE is considered desirable
- D. A higher value of MAE is considered desirable

Answer: B — A higher R-squared is desirable.

Why: *Higher R^2 means more variance explained; lower RMSE/MSE/MAE (error metrics) is better.*